ENN582 Exam Cheatsheet v1: RL & Optimal Control

1 · MPC (Model Predictive Control)

Model $x_{k+1} = f(x_k, u_k)$, $x \in X \subseteq \mathbb{R}^n$, $u \in U \subseteq \mathbb{R}^m$, $f(0,0)=0$. Prediction $x_{k+\ell|k} = f(x_{k+\ell|k}, u_{k+\ell|k})$, $x_{k|k} = x_k$ (measured).

Cost $J(x_k, \vec{u}) = \Sigma_{\ell=0}^{N-1} [x^TQx + u^TRu]_{k+\ell|k} + x^T_{k+N|k} Q_F x_{k+N|k}$, with $Q \geq 0$, $Q_F \geq 0$, $R > 0$. Design knobs: L, F, N.

Q = weight on state error → how hard we push x to 0 (speed/accuracy). **R** = weight on input → control effort/energy, actuator wear. **Q_F** = terminal weight → stands in for cost after the horizon; key to stability. Large Q/R ⇒ fast convergence, big inputs. Small Q/R ⇒ slow, gentle inputs. Diagonal entries weight individual states (e.g. Q = diag(1,0) penalises position only). R = ρI: ρ trades input energy J_{in} vs output energy J_{out}.

Receding horizon: (1) measure x_k; (2) solve $\vec{u}^*_k = \text{argmin}_{\vec{u}\in U^N} J(x_k, \vec{u})$; (3) apply only the first element $u_k = u^*_{k|k} = [I_m\ 0\ \dots\ 0]\ \vec{u}^*_k$; (4) $k \leftarrow k+1$, shift window, repeat. Gives implicit feedback law $u_k = \kappa(x_k)$. Usually $u^*_{k+1|k} \neq u^*_{k+1|k+1}$. Why: feedback corrects disturbances/model errors, handles constraints, looks ahead, finite computation. Longer N ⇒ better but more computation; moderate N ≈ infinite-horizon performance.

Batch form (LTI) $x_{k+\ell|k} = A^\ell x_k + \Sigma_{j=0}^{\ell-1} A^{\ell-j-1}Bu_{k+j|k}$; $\vec{x}_k = \Phi\vec{u} + \Lambda x_k$, $\Phi = [B\ 0\ \dots\ 0;\ AB\ B\ \dots\ 0;\ \dots;\ A^{N-1}B\ \dots\ AB\ B]$, $\Lambda = [A; A^2; \dots; A^N]$, $Q = \text{diag}(Q,\dots,Q,Q_F)$.
$J = x_k^T(\Lambda^TQ\Lambda+Q)x_k + \vec{u}^T(\Phi^TQ\Phi+\rho I)\vec{u} + 2\vec{u}^T\Phi^TQ\Lambda x_k$
$\vec{u}^* = -(\Phi^T\bar{Q}\Phi + \rho I)^{-1}\Phi^T\bar{Q}\Lambda\, x_k$; $u_k = -Kx_k$, $K = [I\ 0\dots0](\Phi^TQ\Phi+\rho I)^{-1}\Phi^TQ\Lambda$ = finite-horizon LQR. $\partial(u^TXu)/\partial u = 2Xu$, $\partial(a^Tu)/\partial u = a$.

Constraints → QP: $\min \vec{u}^TH\vec{u} + \vec{u}^Tf$ s.t. $A_{in}\vec{u} \leq B_{in}$ (may be infeasible). Integral action: penalise $\Delta u = u_{k+\ell+1} - u_{k+\ell}$. Double integrator: $A = [1\ \Delta t;\ 0\ 1]$, $B = [\Delta t^2/2;\ \Delta t]$. ZOH: $[A\ B;\ 0\ I] = \exp([A_c\ B_c;\ 0\ 0]T_s)$.

Stability (opt.) A1: $F(0)=0$, $F \geq 0$, $L(0,0)=0$, $L(x,u) \geq \alpha(\|x\|)$. A2: $\exists \kappa_f$: $F(f(x,\kappa_f(x))) \leq F(x) - L(x,\kappa_f(x))$. Shifted sequence $[u^*_{k+1|k},\dots,\kappa_f(x^*_{k+N|k})] \Rightarrow 0 \leq V_{k+1} \leq V_k - \alpha(\|x_k\|) \Rightarrow x_k \to 0$. Finite input set: R=0, κ_f=0, Q_F from Lyapunov $A^TQ_FA + Q = Q_F$ (A stable; scalar $Q_F = q/(1-a^2)$).

Platoon $e_{i,k} = x_{i-1,k} - x_{i,k}$, $e_{k+1} = Ae_k + B\bar{u}_k + w_k$, $u_i = u_{i-1} - \bar{u}_i$. DARE: $S = A^TSA - A^TSB(R+B^TSB)^{-1}B^TSA + Q$, $K = (R+B^TSB)^{-1}B^TSA$. LQR ignores constraints (crash, 288 km/h); MPC adds p_{min}, v_{max}, a, u limits.

2 · Finite-horizon DP (deterministic)

Principle of optimality: the tail of an optimal trajectory is optimal for the tail subproblem starting from that point, no matter how it was reached.

$J(x_0,\vec{u}) = g_N(x_N) + \Sigma_{k=0}^{N-1} g_k(x_k,u_k)$. **DP**: $J_N(x) = g_N(x)$; $J_k(x) = \min_{u\in U}\{g_k(x,u) + J_{k+1}(f_k(x,u))\}$, k = N−1…0 (backward, all states). $J^* = J_0(x_0)$. Forward pass: $u^*_k = \text{argmin}\{\dots\}$, $x^*_{k+1} = f(x^*_k,u^*_k)$. Curse of dimensionality: needs every state ⇒ discretise.

Shortest path: dist(E)=0, dist(X)=min over edges {c(X→Y)+dist(Y)}. Notes: S→C→D→E, cost 6.

LQR by DP: $S_N = Q_F$; $S_k = A^TS_{k+1}A + Q - A^TS_{k+1}B(R+B^TS_{k+1}B)^{-1}B^TS_{k+1}A$; $L_k = -(R+B^TS_{k+1}B)^{-1}B^TS_{k+1}A$; $u^*_k = L_kx_k$; $J^* = x_0^TS_0x_0$, $J_k(x) = x^TS_kx$.
Scalar (a,b,q,r): $S_k = q + a^2rS_{k+1}/(r + b^2S_{k+1})$, $L_k = -abS_{k+1}/(r + b^2S_{k+1})$.

3 · Stochastic DP & MDPs

$x_{k+1} = f_k(x_k,u_k,w_k)$. Optimise over **policies** $\pi = \{\mu_0,\dots,\mu_{N-1}\}$, $u_k = \mu_k(x_k)$ (use future state info ⇒ beat open-loop sequences). $J_\pi(x_0) = E\{g_N(x_N) + \Sigma\, g_k(x_k,\mu_k(x_k),w_k)\}$.
DP: $J_k(x) = \min_u E_w\{g_k(x,u,w) + J_{k+1}(f_k(x,u,w))\}$.

Expectation tricks (E w=0, E $w^2=\sigma^2$, E[xw]=E[uw]=0): $E[(a+w)^2] = a^2 + \sigma^2$; E[c]=c; E[βw]=0. Vector: $E[w^TSw] = \text{tr}(S\Sigma)$.

Exam-style template $x_{k+1} = x+u+w$, $g_k = g_N = \beta x^2$: $J_k(x) = \beta x^2 + (N-k)\beta\sigma^2$, $\mu_k(x) = -x$, $J_0 = \beta(x_0^2 + N\sigma^2)$. N=3: $J_2 = \beta x^2+\beta\sigma^2$, $J_1 = \beta x^2+2\beta\sigma^2$, $J_0 = \beta x^2+3\beta\sigma^2$; J_0=1 ⇔ $\beta = 1/(x_0^2+3\sigma^2)$.

LQR + additive noise: same gains L_k (certainty equivalence); $J_0 = x_0^TS_0x_0 + \Sigma_{k=0}^{N-1} E[w_k^TS_{k+1}w_k]$.
Dropouts $B_k = B$ w.p. Γ, 0 w.p. 1−Γ: $L_k = -(R+\Gamma B^TS_{k+1}B)^{-1}\Gamma B^TS_{k+1}A$; $S_k = A^TS_{k+1}A + Q - A^TS_{k+1}\Gamma B(R+\Gamma B^TS_{k+1}B)^{-1}\Gamma B^TS_{k+1}A$.

MDP: $p_{ij}(u) = P(x_{k+1}=j \mid x_k=i, u_k=u)$, rows sum to 1. $J_k(i) = \min_u\{g(i,u) + \Sigma_j\, p_{ij}(u)J_{k+1}(j)\}$. Equivalent to $x_{k+1} = w_k$.

Rover X={T,R,B}, U={0 idle, 1 drive}. P(0) = [.75 .25 0; 0 0 1; 0 0 1], P(1) = [.8 .2 0; .9 0 .1; 0 .1 .9]. g(T,0)=−3, g(T,1)=−1, g(R,0)=g(B,0)=0, g(R,1)=g(B,1)=2. N=3, g_3=0: J_2=(−3,0,0); J_1=(−5.25,−0.7,0); J_0=(−7.1125,−2.725,0). Policy: T→0 always; R→1 at k=0,1, 0 at k=2; B→0.

4 · Infinite horizon: VI & PI

$J_\pi(x) = \lim E\{\Sigma_k\, \alpha^k g(x_k,\mu_k(x_k),w_k)\}$, $\alpha \in [0,1)$. α=0 myopic; α→1 far-sighted (slower convergence). Stationary policies. $J_\pi(x_0) = E\{g(x_0,\dots)\} + \alpha J_\pi(x_1)$. $\Sigma_{k\geq0}\alpha^k = 1/(1-\alpha)$; $\Sigma_{k\geq K}\alpha^k = \alpha^K/(1-\alpha)$.

Bellman: $J^*(x) = \min_u E\{g(x,u,w) + \alpha J^*(f(x,u,w))\}$; finite: $J^*(i) = \min_u\{g(i,u) + \alpha\Sigma_j p_{ij}(u)J^*(j)\}$. Operator $TJ = \min_u E\{g+\alpha J(f)\}$; $J^* = \lim T^kJ$, error ≤ $\alpha^KM/(1-\alpha)$.

VI: V_0 = any bounded (0); $V_{k+1}(i) = \min_u\{g(i,u) + \alpha\Sigma_j p_{ij}(u)V_k(j)\}$. Matrix: $V_{k+1} = \min\{G(0)+\alpha P(0)V_k,\ G(1)+\alpha P(1)V_k\}$ (componentwise). V_k = optimal k-stage cost; policy = argmin. Policy often settles before values (not guaranteed).

PI: start μ^0. (1) Evaluate: $J_\mu = g_\mu + \alpha P_\mu J_\mu \Rightarrow J_\mu = (I - \alpha P_\mu)^{-1}g_\mu$, or iterate $J_{\mu,k+1} = g_\mu + \alpha P_\mu J_{\mu,k}$. (2) Improve: $\mu^{\ell+1}(i) = \text{argmin}_u\{g(i,u) + \alpha\Sigma p_{ij}(u)J_{\mu^\ell}(j)\}$. Stop when μ unchanged. Finite #policies ⇒ finite steps. Actor (improve) – critic (evaluate).
2×2 inverse: $[a\ b;\ c\ d]^{-1} = (1/(ad-bc))[d\ -b;\ -c\ a]$.

VI vs PI: VI cheap per iteration, many iterations, values converge asymptotically. PI: expensive evaluation (linear solve), few iterations, exact policy in finite steps. Variants: truncated evaluation (optimistic PI), asynchronous updates (only visited states). Rover: α=0.9 ⇒ μ=(0,1,0); α=0.96 ⇒ μ=(0,1,1) (drive at bottom too). Average cost: lim (1/N)E Σg.

5 · Approximate DP

Value-space approx.: $\tilde{u}_k \in \text{argmin}_u\{g_k(x_k,u) + \tilde{J}_{k+1}(f_k(x_k,u))\}$, forward only, no backward pass. $\tilde{J} = J \Rightarrow$ optimal.

Rollout: $\tilde{J}_{k+1}$ = cost of running a base (heuristic) policy from x_{k+1} to the end. Policy improvement property: rollout cost ≤ base cost. TSP (A start): AB100 AC300 AD100 AE75 BC50 BD75 BE125 CD100 CE125 DE50. Nearest neighbour gives AEDBCA = 550; rollout with N.N. base: stage 1 B(375) C(550) D(625) E(550) → B; stage 2 C375 D650 E675 → C; stage 3 D375 E425 → D ⇒ ABCDEA = 375 (optimal here, not always).

Stochastic: $\tilde{J}$ approx + approximate E. **Certainty-equivalent control**: replace w_k by a single $\bar{w}_k$ (usually E w_k) ⇒ deterministic problem; optimal for LQG.
ℓ-step lookahead: min over u_k, $\mu_{k+1}\dots\mu_{k+\ell-1}$ of $E\{\Sigma\, g + \tilde{J}_{k+\ell}\}$; apply first control only (like MPC). Larger ℓ compensates poor $\tilde{J}$, costs more. **Truncated rollout**: base policy for ℓ−1 steps + terminal $\tilde{J}$ (often $\tilde{J}$=0 fine if ℓ large). Infinite: $\tilde{\mu}(x) \in \text{argmin}\ E\{g + \alpha\tilde{J}(f)\}$.

Parametric $\tilde{J}(x,r)$; features φ(x) (handcrafted, aggregation); linear $\tilde{J} = r^T\varphi(x)$ (r = weights). Train on (x^s, β^s): $\min \Phi(r) = \tfrac{1}{2}\Sigma_s(\tilde{J}(x^s,r) - \beta^s)^2$. Linear ⇒ convex: $\hat{r} = (\Sigma\varphi\varphi^T)^{-1}\Sigma\beta^s\varphi$ (least squares). GD: $r^{\ell+1} = r^\ell - \gamma^\ell\Sigma_s\varphi(x^s)(r^{\ell T}\varphi(x^s) - \beta^s)$; step γ decreasing; SGD = one random sample per step. Finite horizon: fit r_k backwards for each k.

6 · Reinforcement learning (model-free, tabular)

Agent applies u, observes cost g and next state x′; no model f. Alternative: system ID then MPC/DP.

TD(0) prediction of J_μ: $J(x_k) \leftarrow J(x_k) + \delta_k[g + \alpha J(x_{k+1}) - J(x_k)]$. TD target = $g + \alpha J(x_{k+1})$; TD error = target − $J(x_k)$. Robbins–Monro stochastic approximation.

Q-factors: $Q_\mu(x,u) = E\{g(x,u,w) + \alpha J_\mu(f(x,u,w))\}$; $J^*(x) = \min_u Q^*(x,u)$; $\mu^*(x) = \text{argmin}_u Q^*(x,u)$. Q-Bellman: $Q^*(x,u) = E\{g + \alpha$

$\min_{u'} Q^*(f(x,u,w),u')\}$.

Sarsa (on-policy): $Q(x,u) \leftarrow Q(x,u) + \delta[g + \alpha Q(x',u') - Q(x,u)]$, u' = action actually taken next (ε-greedy). Data (x,u,g,x',u') = S,A,R,S,A.

Q-learning (off-policy): $Q(x,u) \leftarrow Q(x,u) + \delta[g + \alpha \min_{u'} Q(x',u') - Q(x,u)]$. Learns greedy Q* while acting ε-greedy.

ε-greedy: greedy argmin Q w.p. $1-\varepsilon$, random $u \in U$ w.p. $\varepsilon \Rightarrow$ $P(\text{greedy}) = 1-\varepsilon+\varepsilon/|U|$. Exploration vs exploitation. ε_k decreasing, e.g. 1/k or max(1/k, 0.01).

Convergence: all (x,u) visited infinitely often, $\delta_k>0$, $\Sigma\delta_k=\infty$, $\Sigma\delta_k^2<\infty$ ($\delta_k=1/k$ ✓, constant δ ✗, $1/k^2$ ✗), $\varepsilon \to 0$ (greedy in limit). Constant ε $\Rightarrow$ Sarsa converges to Q of the ε-greedy policy. Terminal-state Q = 0 initially.

Rewards instead of costs: maximise (use max, argmax), or cost = −reward. Deep RL: $Q(x,u;r)$, minimise $E[(g + \alpha \min Q(x',\cdot;r) - Q(x,u;r))^2]$ by GD. Learning ≠ guaranteed closed-loop stability.

Hand-calc table: step | (x,u) | g | (x',u') | Q_{old} | target $g+\alpha Q(x',u')$ | $Q_{new} = Q_{old} + \delta(\text{target} - Q_{old})$. Always use the current table value (updates from earlier steps/episodes carry forward).

7 · Answer recipes

Stochastic DP by hand: (1) $J_N = g_N$. (2) Substitute $x' = f(x,u,w)$ into J_{k+1}. (3) Expand squares, take E (w-terms: E w = 0, E $w^2 = \sigma^2$). (4) $d/du = 0 \to u^*$; check 2nd derivative > 0. (5) Plug u^* back → J_k. (6) Repeat to k=0. State policy $\mu_k(x)$ and $J_0(x_0)$.

Finite MDP DP / VI step: for each state i and each u: $g(i,u) + \alpha\Sigma_j p_{ij}(u)V(j)$ (α=1 for finite-horizon DP). Take min, record argmin. Show a table: state | u=0 value | u=1 value | min | policy.

PI step: write linear eqs $J(i) = g(i,\mu(i)) + \alpha\Sigma p_{ij}(\mu(i))J(j)$; solve; then improvement table as above; stop when policy repeats.

Sarsa / Q-learning by hand: list transitions from episodes (x,u,r,x',u'); table of all Q = 0; update one transition at a time in order; carry the table into the next episode; final table; greedy policy (min for cost, max for reward); explain ties (one-step lookahead r + α·best Q(next), or note more episodes needed).

MPC by hand (scalar): N=1: $J = qx^2 + ru^2 + q_F(ax+bu)^2$, $u^* = -abq_Fx/(r + b^2q_F)$. N≥2: Riccati backwards from $S_N = q_F$, use L_0 only (receding horizon); closed loop $x_{k+1} = (a + bL_0)x_k$, stable iff $|a + bL_0| < 1$.

Key words to use: principle of optimality, cost-to-go, tail subproblem, Bellman equation, stationary policy, contraction, policy evaluation/improvement, base policy, cost improvement, certainty equivalence, features/weights, TD target/error, on/off-policy, exploration–exploitation, implicit feedback law κ(x), constraints, QP, feasibility, terminal cost, Lyapunov.

8 · 2024 exam answers (quick)

Q1: (a) Q state penalty → speed of regulation; R input penalty → effort; Q_F terminal → tail cost/stability; ratio Q:R sets aggressiveness. (b) Plan N steps, apply first, re-measure, re-plan (moving window).

Q2: $\mu_k(x) = -x$ (k=0,1,2); $J_0 = \beta x_0^2 + 3\beta\sigma^2$; $\beta = 1/(x_0^2 + 3\sigma^2)$.

Q3 (δ=0.1, α=0.5, Sarsa): ep1: Q(1,A)=0, Q(2,A)=0.1·(100+0)=10, Q(4,B)=0.1·(100+0.5·0)=10. ep2: Q(1,B)=0, Q(3,A)=0.1·(0+0.5·10)=0.5, Q(2,A)=10+0.1(100+0.5·10−10)=19.5. Final: Q(1,A)=0, Q(1,B)=0, Q(2,A)=19.5, Q(3,A)=0.5, Q(4,B)=10, rest 0. Rewards $\Rightarrow$ argmax. Tie at 1: break by r + α·max Q(next): A → 0.5·19.5 = 9.75 > B → 0.5·0.5 = 0.25 $\Rightarrow$ 1 —A→ 2 —A→ 4 (then B self-loop).

9 · Practicals P1–P6: results & lessons

P1 crane $x=[p\ v\ \theta\ \omega]$, $x_0=[2;0;0;0]$, $x^*=[10;0;0;0]$; $0\le p\le 12$, $|v|\le 3$, $|\theta| \le 10°=0.1745$, $|\omega|\le 50°/s=0.8727$, $|u|\le 100$ N, $T_s=0.1$, ZOH $[A\ B;0\ I]=e^{[A_c\ B_c;0\ 0]T_s}$. Bryson $Q_{ii}=1/x_{max}^2$, $R=1/u_{max}^2$ → diag(.016,.11,32.8,1.31), 10^{-4}. LQR diag(.5,1,500,50), R=1: 7.3 s (conservative: biggest push at t=0; limits met only by tuning). MPC Q=I, R=1, N=30: 5.4 s on limits. QP: $H=2(\Phi^TQ\Phi+R)$, $f=2\Phi^TQ(\Lambda x_k-\bar{x}^*)$, $A_{in}=[-\Phi;\Phi;-I;I]$, $b_{in}=[\Lambda x_k-x_{min};\ x_{max}-\Lambda x_k;\ -u_{min};\ u_{max}]$; N vars, 10N rows. Short N → infeasible; limits only at samples.

P2 J(E,F,G)=807,450,603; J(B,C,D)=730,854,1056; J(A)=1063 A-B-F-H; greedy A-D-E-H=1305. TSP: 13 (A-B-D-C-A); state=(city, unvisited set); $(n-1)!/2$ tours vs $n^2 2^n$. NN = 1-step lookahead, J=0. AB=50: NN 58, opt 40 (A-C-B-D-A), rollout(NN) finds 40, never worse than base.

P3 E{g} per edge = P2 costs $\Rightarrow$ same 1063 (noise only in cost $\Rightarrow$ certainty equivalence exact). Fork: J(B,C,D,E)=883,920,718,973; East .6(333+883)+.4(282+920)=1210.4; West .8(230+718)+.2(300+973)=1013 $\Rightarrow$ West. Parking: J_N=S; $J_k(\text{full})=c+\rho J_{k+1}(\text{free})+(1-\rho)J_{k+1}(\text{full})$; $J_k(\text{free})=\min(s(k), J_k(\text{full}))$. $c \ge$ fee drop $\Rightarrow$ park at first free; else threshold rule.

P4 Grid VI: $q=r(\text{next})-1+\alpha V(\text{next})$, α=.9, V(0,0)=−4.0951; hole reset $\Rightarrow$ bootstrap αV(start). Frozen lake (slip: intended or ⊥, ⅓ each): det V(0)=$.99^5$=.951; slippery V(0)=.542; det policy on ice 4.7% vs 74.7%; walls used as protection. VI 180 its vs PI 5; policy final at VI it 45. PI: keep old action on ties. Exact PE $(I-\alpha P_\mu)V=r_\mu$.

P5 $\tilde{\mu}(x)=\text{argmax}[r-1+J(f(x,u))]$, J=−Euclid $\Rightarrow$ loop (3,0)↔(3,1); ℓ=3 optimal; cost $\propto b^\ell$. φ=normalize$[x_1,x_2,|x_1-4|+|x_2-4|]$ (sum ≡ 8), r=[10,10,2] exact. SGD $r \leftarrow r-\gamma\varphi(r^T\varphi-\beta)$, $\beta=\max_u[-1+r^T\varphi(\text{next})]$; e.g. (1,1): φ=[.125,.125,.75], β=−1 $\Rightarrow$ r=−.01φ.

P6 Cliff 4×12, −1/step, −100 fall. Q-learning (max) $\Rightarrow$ 13-step edge path −13. SARSA (Q(x′,u′), ε-moves included) $\Rightarrow$ 15/17-step safe detour; ε=0 $\Rightarrow$ same as Q-learning. Resetting k each episode $\Rightarrow$ ε never → 0, Robbins–Monro broken. Edge example: Q=−2, r=−1, Q(x′,·)={−1, −100}, u′=down, δ=.5: SARSA −51, QL −1.995.